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A Review of Robust Operations Management under Model Uncertainty

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Over the past two decades, there has been explosive growth in the application of robust optimization in operations management (robust OM), fueled by both significant advances in optimization theory and a volatile business environment that has led to rising concerns about model uncertainty. We review some common modeling frameworks in robust OM, including the representation of uncertainty and the decision-making criteria, and sources of model uncertainty that have arisen in the literature, such as demand, supply, and preference. We discuss the successes of robust OM in addressing model uncertainty, enriching decision criteria, generating structural results, and facilitating computation. We also discuss several future research opportunities and challenges.

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1. Introduction

Traditional operations management (OM) models typically require complete model information, including deterministic parameters that are precisely known, random parameters that have known probability distributions, and dependence relationships (e.g., how demand depends on price) that are characterized by known functions. However, in practice, *model uncertainty* or *ambiguity* is ubiquitous. Deterministic parameters may suffer from estimation errors and data contamination. Random parameters may have ambiguous distributions, with only partial distributional information available, such as support, mean, and variance. Both the functional form and the parameters of dependence relationships can be unknown. Solving OM models with model uncertainty requires nontraditional methods, one of which is robust optimization.

OM models that would now be classified as robust or distributionally robust optimization date back to the late 1950s and early 1960s. General robust optimization models were investigated as early as the 1970s. From the mid-1990s to the mid-2000s, robust optimization models were revitalized by a series of breakthroughs in theory and methodology, which soon prompted vast applications in both traditional OM areas, such as inventory management and production planning, pricing and revenue management, scheduling and project management, transportation and vehicle routing, and facility location and network design, and emerging OM areas such as health care, humanitarian operations, and sustainability. The application of robust optimization remains one of the most active and rapidly developing areas in the study of OM.

We review applications of robust optimization in operations management, that is, research that we refer to as *robust OM*. In contrast to existing reviews that categorize the literature according to specific applications, we focus on the sources of uncertainty — such as demand and supply—and the modeling framework, including the representation of the uncertainty and the decision criteria, because we believe they better represent the essence of the research. By doing so, we seek to account for the successes of robust OM in addressing model uncertainty, enabling flexible objectives, generating structural results, and facilitating computation. We then discuss several possible research opportunities and challenges, including the interplay between robust OM and topics such as data science, operations in emerging areas, and stochastic orders, and more efficient solution methods for computation-oriented robust OM models.

The remainder of the study is organized as follows. In section 2, we briefly review common modeling frameworks in robust optimization and discuss examples of their applications in OM. In section 3, we review different sources of uncertainty in robust OM and discuss how they are represented using uncertainty sets. In sections 4 and 5, we discuss several reasons for the successes of robust OM and outline some future research opportunities. It is not our intent to provide a comprehensive review of the theory of robust optimization or its applications in OM, for which we refer the reader to excellent books and surveys by Ben-Tal et al. (2009a), Bertsimas et al. (2011), Gabrel et al. (2014), Gorissen et al. (2015), Kouvelis and Yu (1997), Lim et al. (2006), Rahimian and Mehrotra (2019), Sözüer and Thiele (2016), and Yanıkoğlu et al. (2017).

2. Modeling Framework and Solution Approach

In this section, we briefly introduce elements of the robust optimization modeling framework and their applications in OM. To facilitate the introduction, we use the newsvendor problem as an example. Consider a retailer of n items. For item $i = 1,2,\ldots,n$, let d_i, s_i, and c_i denote the demand, unit selling price, and unit cost, respectively. Given the exact demand d_i and order quantity q_i, the retailer's profit from item i is given by $\pi_i(q_i, d_i) = s_i \min\{q_i, d_i\} - cq_i$. The demand is uncertain, and the retailer must determine the order quantities q_i, $i = 1,2,\ldots,n$, before knowing the exact demand. The order quantities could be interdependent, for example, through a capacity constraint $\sum_i b_i q_i \leq \bar{b}$. Consider the single-item case (i.e., $n = 1$), in which the subscript i in the notation is omitted. Traditionally, demand uncertainty is represented by a random variable D with a known probability distribution P, and the optimal order quantity q is chosen to maximize the expected profit. Such *uncertainty representation* and *decision criterion* lead to the classical newsvendor model

$$\max_q \mathbb{E}_P[\pi(q, D)].$$

Robust optimization provides alternative uncertainty representation and decision criteria for OM problems under uncertainty. Robust versions of the (multi-item) newsvendor problem have been studied by Ardestani-Jaafari and Delage (2016), Gallego and Moon (1993), Hanasusanto et al. (2015), Moon and Silver (2000), Scarf (1957), Vairaktarakis (2000a), Wiesemann et al. (2014).

2.1. Uncertainty Representation

Representing uncertainty as deterministic sets of parameters instead of probability distributions marks the difference between robust and stochastic optimization. Uncertainty sets must be chosen appropriately to provide sufficient protection against unexpected outcomes while controlling the level of conservatism, and to utilize available information effectively while keeping the resultant problem tractable or easy to approximate.

2.1.1. Parameter Uncertainty. In the newsvendor problem, assume the demand is in an uncertainty set $\mathcal{U}$. Depending on the available information (e.g., forecasts, expert opinion), different forms of $\mathcal{U}$ can be constructed. When the demand is known to have m possible values, one can use a *finite scenario* uncertainty set

$$\mathcal{U} = \{d^1, d^2, \ldots, d^m\}.$$

Scenario uncertainty sets were considered by Mulvey et al. (1995), who used the term "robust optimization" for mathematical programming models with uncertain parameters. The book Kouvelis and Yu (1997) and a series of papers by them and their coauthors dealt with this type of uncertainty for various OM problems such as inventory, location, and routing.

In practice, it is common to have estimates or forecasts with confidence or prediction intervals. This type of information motivates the use of *interval* or *box* uncertainty sets, where each uncertain parameter lies within a known interval. For example, in the multi-item newsvendor problem, the following uncertainty set can be constructed for demand $\mathbf{d} = (d_1, d_2, \ldots, d_n)$,

$$\mathcal{U} = \left\{ \mathbf{d} \in \mathbb{R}^n_+ | d_i \in [\bar{d}_i - \hat{d}_i, \bar{d}_i + \hat{d}_i], \forall i \right\},$$

where $\bar{d}_i$ is the demand forecast (i.e., the nominal value) and $\hat{d}_i$ is the forecast error bound (i.e., the maximum deviation from the nominal value) for item i. Interval uncertainty was considered for inventory management by Kasugai and Kasegai (1960), and later for scheduling and location by Kouvelis and Yu (1997). It has the advantage of having tractable reformulations for many OM problems and continues to be a common form of uncertainty in the robust OM literature. However, as Ben-Tal and Nemirovski (1999) and Bertsimas and Sim (2004) pointed out, robust optimization with interval uncertainty may be overly conservative, since it is not likely for multiple uncertain parameters to take on their extreme values simultaneously.

To address the over-conservatism of the interval uncertainty set, the amount of deviation from the nominal value needs to be controlled. Ben-Tal and Nemirovski (1999) proposed the *ellipsoid* uncertainty set,

$$\mathcal{U}=\left\{\mathbf{d}\in\mathbb{R}_+^n|d_i\in[\bar{d}_i-\hat{d}_i,\bar{d}_i+\hat{d}_i],\forall i,\sqrt{\sum_i\left(\frac{d_i-\bar{d}_i}{\hat{d}_i}\right)^2}\le\rho\right\},$$

where the parameter ρ controls the level of conservatism by restricting the demand within an ellipsoid around the nominal value. Similarly, one can construct a *polytope* uncertainty set with linear inequality constraints, which can be viewed as a special case of the ellipsoid uncertainty set, as pointed out by Ben-Tal and Nemirovski (1999). Robust linear programs with ellipsoid uncertainty sets have tractable reformulations as conic quadratic programs (Ben-Tal et al. 2009a). It has been applied to the study of various robust OM problems, such as supply chain contracting (Ben-Tal et al. 2005), network design (Mudchanatongsuk et al. 2008), and facility location (Baron et al. 2011).

Bertsimas and Sim (2004) proposed the uncertainty set

$$\mathcal{U}=\left\{\mathbf{d}\in\mathbb{R}_+^n|d_i\in[\bar{d}_i-\hat{d}_i,\bar{d}_i+\hat{d}_i],\forall i,\sum_i\frac{|d_i-\bar{d}_i|}{\hat{d}_i}\le\Gamma\right\},$$

where the parameter $\Gamma < n$, called the *budget of uncertainty*, controls the level of conservatism. In their robust optimization model, the parameter Γ limits the number of items whose demand can deviate from the nominal value. Therefore, this uncertainty set is also known as the *cardinality-constrained* uncertainty set. Robust linear programs with the cardinality-constrained uncertainty set can be reformulated as linear programs. It has been widely applied to robust OM problems, including inventory management (Bertsimas and Thiele 2006b, Bienstock and Ozbay 2008), network design (Atamtürk and Zhang 2007), energy systems (Jiang et al. 2012), and project management (Bruni et al. 2017).

2.1.2. Distribution Ambiguity. When certain incomplete distribution information is available, uncertainty can be represented using ambiguous probability distributions. For example, the demand distribution P may not be completely known, but is believed to be in an *ambiguity set* $\mathcal{A}$ of possible probability distributions. Depending on what type of distribution information is available, different forms of ambiguity sets can be constructed. When the demand mean μ and variance σ^2 are known, the *moment* model is based on the following ambiguity set,

$$\mathcal{A}=\left\{P\in\mathcal{P}|\mathbb{E}_P[D]=\mu,\mathbb{E}_P[D^2]=\mu^2+\sigma^2\right\},\qquad((1))$$

where $\mathcal{P}$ is the set of all demand distributions. Support information, covariance, and uncertain moments can also be incorporated (Delage and Ye 2010). The moment model for the newsvendor problem was first investigated by Scarf (1957) and later by Gallego and Moon (1993) with extensions. Fueled by research advances in distributionally robust optimization and conic optimization, the application of moment models in OM has witnessed an explosive growth in recent years, including choice modeling (Natarajan et al. 2009b), infrastructure planning (Mak et al. 2013), appointment scheduling (Kong et al. 2013, 2020, Mak et al. 2015, Zhang et al. 2017b), hospital admission (Meng et al. 2015), supply chain contracting (Fu et al. 2018, Wagner 2015), process flexibility (Wang and Zhang 2015), workforce scheduling (Yan et al. 2017), vehicle sharing (Hao et al. 2019, He et al. 2017, 2020), closed-loop supply chains (Jiao et al. 2018), supply chain risk mitigation (Gao et al. 2019), and last-mile delivery (Liu et al. 2020).

Another form of ambiguity arises when only the marginals of a joint distribution are known. In the multi-item newsvendor problem, assume the demand for item i has a known marginal distribution P_i, while the joint distribution of $\mathbf{D}=(D_1,D_2,\ldots,D_n)$ is unknown. In this case, the ambiguity set $\mathcal{A}=\mathcal{P}(P_1,P_2,\ldots,P_n)$ contains all joint distributions with marginals $P_1,P_2,\ldots,P_n$. Such distribution ambiguity has been considered for task completion times in project management (Meilijson and Nádas 1979), consumer utility in discrete choice models (Mishra et al. 2014, Natarajan et al. 2009b, Yan et al. 2016), dependent supply and demand in supply chains (Mak and Shen 2014), disruption risks in facility location (Lu et al. 2015), random capacity in multi-sourcing (Zhao and Freeman 2016, 2018), and service times in appointment scheduling (Chen et al. 2018a). In many of these applications, marginal distribution models are used to uncover the impact of correlation or dependence, also known as the price of correlation (Agrawal et al. 2012).

When the actual distribution is believed to be close to a known nominal distribution (e.g., a distribution fitted from historical data), an ambiguity set can be constructed using divergence measures or metrics between distributions. For example, when the nominal distribution Q and the actual distribution P have probability density function f_Q and f_P, respectively, the *Kullback-Leibler divergence* (also known as *relative entropy*) from Q to P is given by

$$I_{\mathrm{KL}}(P,Q)=\int f_P(x)\log\left(\frac{f_P(x)}{f_Q(x)}\right)\mathrm{d}x.$$

One can construct the following divergence-constrained ambiguity set

$$\mathcal{A} = \{P \in \mathcal{P} | I_{\mathrm{KL}}(P, Q) \leq \rho\},$$

where the parameter ρ controls the degree of ambiguity. Ambiguity sets based on other divergence measures and distance metrics, including special cases or the entire class of ϕ-divergence, the Prokhorov metric, and the Wasserstein metric, have been investigated by Ben-Tal et al. (2013), Erdoğan and Iyengar (2006), Esfahani and Kuhn (2017), Hu and Hong (2013), Jiang and Guan (2016), Klabjan et al. (2013), Lim et al. (2006), Love and Bayraksan (2015), Wang et al. (2016), and Yanıkoğlu and den Hertog (2013). Most of these approaches have been applied to inventory management, which is commonly used as an example to illustrate the efficacy of the proposed ambiguity sets. Other applications in OM include dynamic pricing (Lim and Shanthikumar 2007), queueing systems (Jain et al. 2010), and surgery scheduling (Deng et al. 2019).

The uncertainty and ambiguity sets mentioned above can be refined to include other available information. For example, Chen et al. (2007), Li et al. (2019), Natarajan et al. (2017), and Wiesemann et al. (2014) proposed uncertainty and ambiguity sets that incorporate various statistics and distribution information such as forward/backward divergence, semivariance, higher-order moments, variability measures, and unimodality. Bienstock and Ozbay (2008) considered interval demand uncertainty over multiple periods, combined with a limited number of peak-demand periods. Hanasusanto et al. (2015) considered demand distributions that are mixtures of multiple ambiguous distributions. Chen et al. (2018b) proposed the *scenario-wise* ambiguity set, which can express a wide range of distribution information and has been applied to carsharing fleet management (Hao et al. 2019). Bertsimas et al. (2018a), Delage and Ye (2010), Esfahani and Kuhn (2017), and Jiang and Guan (2016) discussed how to construct various uncertainty and ambiguity sets from data for data-driven robust optimization.

2.2. Decision Criterion

With parameter uncertainty or distribution ambiguity, new notions of feasibility and optimality must be defined. Three criteria have been widely used (Kouvelis and Yu 1997): absolute robustness (optimizing worst-case objective subject to robust feasibility), robust deviation, (also known as min–max regret), and relative robustness (also known as max–min competitive ratio).

2.2.1. Robust Feasibility. In the newsvendor problem with demand $d \in \mathcal{U}$, assume there is an additional requirement that the inventory shortage cannot exceed k. A feasible order quantity q must satisfy this requirement for all possible demand outcomes in the uncertainty set; that is

$$d - q \leq k, \quad \forall d \in \mathcal{U}.$$

Robust feasibility was considered by Soyster (1973) for convex programming under parameter uncertainty. It is particularly important in many OM problems, such as aircraft and crew scheduling (Mulvey et al. 1995) and public service planning (Berman et al. 2013) where the feasibility of decisions under uncertainty is a major concern.

Robust feasibility can also be formulated with respect to ambiguous distributions. For example, when the demand distribution P is in an ambiguity set $\mathcal{A}$, the decision-maker may specify that the expected shortage is less than or equal to k under any possible demand distribution. Alternatively, it may require that the probability of having no more than k units of shortage is at least $1-\varepsilon$ under any possible demand distribution; that is

$$P(D - q \leq k) \geq 1 - \epsilon, \quad \forall P \in \mathcal{A}.$$

Such a constraint provides a probabilistic guarantee and is known as an *ambiguous chance constraint*. Parameter uncertainty sets can also provide probabilistic guarantee under certain distributional assumptions (e.g., Ben-Tal and Nemirovski 1999, Bertsimas and Sim 2004, Chen et al. 2007). Ambiguous chance-constrained optimization can be applied to OM problems with soft service requirements, such as surgery planning (Deng et al. 2019, Lu et al. 2020, Zhang et al. 2017a) and energy systems planning (Xie and Ahmed 2018, Zhang et al. 2017c).

2.2.2. Worst-Case Objective. To choose from a set of robustly feasible solutions, one can optimize the objective under the worst possible outcome. For example, in his seminal paper, Scarf (1957) studied the newsvendor problem with distribution ambiguity. The objective is to maximize the expected profit under the worst-case distribution in the moment ambiguity set $\mathcal{A}$ defined in Equation (1); that is

$$\max_{q} \min_{P \in \mathcal{A}} \mathbb{E}_P[\pi(q, D)]. \qquad ((2))$$

This criterion is usually referred to as "min–max" for a minimization problem or "max–min" for a maximization problem. A model with both a worst-case objective and robust feasibility constraints was referred to as *absolute robustness* by Kouvelis and Yu (1997).

Technically speaking, a robust optimization model with a worst-case objective can be equivalently formulated as a problem with only robust feasibility constraints, by adding auxiliary variables and corresponding constraints that represent the worst-case objective value. However, as Gabrel et al. (2014) pointed out, "the decision maker generally has different attitudes with respect to infeasibility and sub-optimality." It is therefore necessary to consider the min–max decision criterion and robust feasibility separately. Moreover, the objective of the nominal problem may have multiple equivalent formulations (e.g., with auxiliary variables and constraints). Robust counterparts of equivalent nominal formulations are not necessarily equivalent. The difference between robust feasibility formulations may not be as obvious as the difference between the corresponding max–min formulations, which may result in the use of an unintended decision criterion. Therefore, one should be careful when transforming a nominal problem into its robust counterpart by adding robust feasibility constraints.

In robust OM, the worst-case objective is probably the most widely used decision criterion. There are at least three main reasons for its popularity. First, many min–max or max–min robust optimization models can be solved analytically or efficiently using reformulations. Second, a min–max criterion usually corresponds to a certain risk preference, which is commonly used in traditional OM literature. Third, the worst-case objective may be of special interest to the decision-maker under certain circumstances, such as risk management (Tang 2006).

2.2.3. Min–Max Regret. The min–max (max–min) criterion results in a solution that is optimal under one specific outcome but can be highly suboptimal under other outcomes. The level of suboptimality can be measured by the regret, which is the difference between the objective value and the optimal hindsight objective value given perfect information. The min–max regret approach, also known as *robust deviation* (Kouvelis and Yu 1997), seeks to minimize the maximum regret to control the level of suboptimality under all possible outcomes. For example, the min–max regret newsvendor problem with demand uncertainty $\mathcal{U}$ is given by

$$\min_{q} \max_{d \in \mathcal{U}} \{\pi(d,d) - \pi(q,d)\}.$$

Min–max regret was first applied to inventory management by Kasugai and Kasegai (1961) in response to criticisms that their previous work Kasugai and Kasegai (1960) based on worst-case objective was too conservative. Kouvelis and Yu (1997) investigated a number of robust OM problems with scenario uncertainty. Other applications include location (Averbakh and Berman 1997, 2000a,b), scheduling (Drwal and Rischke 2016, Kasperski 2005a, Lebedev and Averbakh 2006), stochastic inventory management (Levi et al. 2011, Perakis and Roels 2008, Yue et al. 2006), lot sizing (Zhang 2011), energy systems (Jiang et al. 2013), pricing (Caldentey et al. 2017, Chen et al. 2017), and health care (El-Amine et al. 2018).

2.2.4. Max–Min Competitive Ratio. The min–max regret criterion considers the absolute difference between the objective value and the hindsight optimal objective value. If the objective value fluctuates significantly over the uncertainty set, then the magnitude of regret may not truly reflect the level of suboptimality, but is rather driven by the magnitude of the objective value. The competitive ratio, which is the ratio between the objective value and the hindsight optimal objective value, provides a relative measure of optimality, insensitive to the magnitude of the objective value. One can then maximize the minimum competitive ratio to achieve *relative robustness* (Kouvelis and Yu 1997). For example, in the newsvendor problem with demand uncertainty set $\mathcal{U}$, the max–min competitive ratio criterion leads to the following problem

$$\max_{q} \min_{d \in \mathcal{U}} \frac{\pi(q,d)}{\pi(d,d)}.$$

Kouvelis and Yu (1997) examined a number of OM problems under the relative robustness framework. Yu (1997) investigated the relative robust economic order quantity (EOQ) problem. Vairaktarakis (2000b) investigated the multi-item newsvendor problem with interval demand. Zhu et al. (2013) studied the relative robust newsvendor problem with ambiguous distribution to quantify the relative expected value of distribution.

2.2.5. Other Criteria. Several other decision criteria have been proposed to address the overconservatism of robust optimization. Mulvey et al. (1995) proposed balancing "solution robustness" (solution quality) and "model robustness" (feasibility) by minimizing a weighted sum of the objective and the penalty for infeasibility. Fischetti and Monaci (2009) proposed the light robustness model based on Bertsimas and Sim (2004), where a linear penalty for infeasibility is minimized while the objective remains within δ of the optimal nominal objective value. Light robustness has been applied to railway operations, because most of the constraint coefficients in these problems are structural and do not have uncertainty. Ben-Tal et al. (2010) proposed the *soft robust* approach with distribution ambiguity, where the feasibility requirement is relaxed but imposed on a larger ambiguity set, and showed that this approach is connected with convex risk measures. Iancu and Trichakis (2014) pointed out that the worst-case objective may result in a Pareto-dominated solution and proposed optimizing over *Pareto robustly optimal* solutions.

Motivated by the theory of *satisficing*, *goal-driven* optimization seeks to maximize the probability of achieving a certain *aspiration level*. Chen and Sim (2009) considered the *shortfall-aware* aspiration-level criterion, where probability is safely approximated with the use of conditional value-at-risk (CVaR). Similar distributionally robust goal-driven models have been applied to inventory management (Lim and Wang 2017), machine scheduling (Zhang et al. 2018a), and infrastructure planning for innovative business models (Mak et al. 2013). Roy (2010) proposed the (b, w)-robustness criterion, where b is an aspiration level, w is a minimally acceptable payoff, and the objective is to maximize the probability (or the number of scenarios) of achieving the aspiration level b, given that the payoff is guaranteed to be greater than w under all scenarios.

The min–max regret criterion may still be overly conservative, because it focuses only on the worst-case regret. Kouvelis et al. (1992) and Gutierrez et al. (1996) proposed to optimize the nominal objective over solutions with relative regret bounded by p. Later referred to by Snyder and Daskin (2006) as *p-robust*, this criterion was applied to facility location and network design (Peng et al. 2011). Kalaï et al. (2012) observed that the coupling of objectives with respect to scenarios in the p-robust approach may not be necessary, and proposed the *lexicographic α-robust* criterion, where regret is calculated using reordered objectives. Daskin et al. (1997) proposed the *α-reliable minimax regret* criterion, which minimizes the value-at-risk of regret. Chen et al. (2006) proposed the *α-reliable mean-excess regret* criterion, which minimizes the CVaR of regret. Both frameworks have been applied to the study of stochastic facility location problems. Natarajan et al. (2013) also proposed minimizing the CVaR of regret with distribution ambiguity.

2.3. Multistage Problems

Many OM problems involve decision-making in multiple periods or stages. Robust multistage optimization has also been widely applied to OM problems. One could adopt a static model, in which all decisions are made up front without being adapted to new information. Static models generally have much better tractability. They also perform surprisingly well in many cases and are in fact optimal for certain problems (Bertsimas et al. 2015). In contrast, having decisions that are adaptive to the realized uncertainty in each stage generally leads to intractability, even for two-stage problems (Ben-Tal et al. 2004).

The most common approach for multistage robust OM is by using decision rules. Ben-Tal et al. (2004) proposed the *affinely adjustable robust counterpart* using linear decision rules, where decisions depend affinely on the uncertainty. Improvement to and generalization of the affinely adjustable model have been proposed by Chen et al. (2008), Chen and Zhang (2009), Goh and Sim (2010, 2011), and See and Sim (2010). These approaches have been applied to OM problems such as capacity expansion (Mudchanatongsuk et al. 2008), inventory management (Ben-Tal et al. 2009b), dynamic pricing (Adida and Perakis 2010), humanitarian relief (Ben-Tal et al. 2011), project management (Goh and Hall 2013), and facility location (An et al. 2014). Bertsimas et al. (2018b) and Chen et al. (2018b) developed affinely adjustable models for adaptive distributionally robust optimization, which has been applied to fleet repositioning in carsharing (He et al. 2020).

2.4. Tractability and Solution Approach

Many of the modeling frameworks mentioned above result in problems that can be solved efficiently. Some may have analytical results for the optimal solution or the structure of the optimal policy (e.g., Bertsimas and Thiele 2006b, Fu et al. 2018, Gallego and Moon 1993, Scarf 1957, Wagner 2018). Others may have tractable *reformulations* (or approximations) in which the robust counterpart is transformed to a deterministic problem with no uncertainty using duality. Ben-Tal et al. (2009a) and Bertsimas et al. (2011) summarized reformulations of different combinations of nominal problems and uncertainty sets. Ben-Tal et al. (2013), Delage and Ye (2010), Wiesemann et al. (2014), and Esfahani and Kuhn (2017) studied reformulations for various distributionally robust models. An alternative to the reformulation approach is the cutting plane approach, where cutting planes are iteratively added to a relaxed problem with a reduced uncertainty set.

Compared to robust linear optimization, robust combinatorial optimization can be much harder to solve. For example, Kouvelis and Yu (1997) showed the robust counterparts of several polynomially solvable combinatorial optimization problems are NP-hard with finite scenario uncertainty. One approach for tackling robust combinatorial problems is the *persistency model* (Bertsimas et al. 2006, Li et al. 2014, Natarajan et al. 2009b, 2011), which estimates the probability that the hindsight optimal decision is equal to one under distribution ambiguity. In addition to the property of the nominal problem, the choice of decision criterion and uncertainty set can also affect the tractability of the robust counterpart. For example, Averbakh and Lebedev (2005) showed that min–max regret linear programming with interval uncertainty is NP-hard, whereas the same problem is polynomially solvable with finite scenario uncertainty. Bertsimas and Sim (2003) showed that under the cardinality-constrained uncertainty set, the robust counterpart of a polynomially solvable 0–1 discrete optimization problem remains polynomially solvable.

3. Sources of Uncertainty

3.1. Demand

3.1.1. Demand Volume. As in the newsvendor example, demand is the most common source of uncertainty in many OM problems. Ambiguity in demand distribution motivated the seminal work of Scarf (1957). Uncertain demand was considered in several of the earliest OM applications of robust optimization after its revival in the 1990s, including demand rate, which is usually assumed to be deterministic in the nominal models, such as the EOQ model (Yu 1997), and periodical demand, which is usually assumed to be random in the nominal models, such as the newsvendor model (Vairaktarakis 2000b), the stochastic lot-sizing problem (Zhang et al. 2016), and multi-period inventory systems with general supply chain structures (Bertsimas and Thiele 2006b).

3.1.2. Demand Location. Compared with uncertain demand volume, uncertain demand location is less studied in OM. Chan et al. (2017) considered the deployment of defibrillators for cardiac arrests with ambiguous location distribution. The authors discretized the continuous area into subregions, which transformed demand location into multidimensional demand volume. They then proved an error bound for using such a discretization approximation. The result suggests discretization could be a promising method for addressing location uncertainty.

3.1.3. Arrival Process. In traditional queueing literature, the interarrival time is typically assumed to have a known distribution. Uncertain interarrival time (as well as service time) has been investigated by Bandi and Bertsimas (2012), Bandi et al. (2015), and Whitt and You (2017) to establish *robust queueing theory*. Results from these robust queueing models not only reconfirmed classical results from stochastic models but also provided new insights.

3.1.4. Spatial and Inter-Temporal Correlation. The impact of correlation between demands at different physical locations or for different items has been extensively studied in OM, which generated insights to important operations strategies such as inventory pooling and product substitution. Robust optimization provides an alternative approach for capturing demand correlation. For example, Mak and Shen (2014) studied inventory pooling under ambiguous demand distributions with known marginals; Agrawal et al. (2012) investigated the price-of-correlation in facility location with random demand. Compared to spatial correlation, inter-temporal demand correlation was less studied in OM due to the challenges it imposes upon analysis. Robust optimization has been applied to incorporate inter-temporal correlation in service-constrained inventory management (Goh and Sim 2011), stochastic lot-sizing (Zhang et al. 2016), and fleet management (He et al. 2020). Whitt and You (2017) applied robust queueing theory to investigate the impact of dependence between interarrival (and service) times.

3.1.5. Dependence on Price and Supply. Another type of demand uncertainty arises when demand depends on decisions such as price and inventory level, but the exact dependence relationship is unknown. One way to model demand uncertainty in this case is to assume a certain parametric demand function (e.g., demand decreases linearly in price) and then to assume that the parameters lie within an uncertainty set. This approach has been applied in pricing and revenue management (e.g., Adida and Perakis 2006, Perakis and Sood 2006), joint pricing-inventory management (Lim 2013), multi-period inventory management with stock-dependent demand (Lim 2019), and facility location with demand mean and variance that depend on the location decisions (Basciftci et al. 2019). An alternative to demand function is to assume that there exists an uncertain total market size and that the effective demand is determined by the price and customer valuation (Haviv and Randhawa 2014). Lim and Shanthikumar (2007) examined a more general ambiguity model for robust dynamic pricing, where the ambiguity set includes all demand processes with bounded relative entropy to a nominal process. A more recent approach proposed by Fu et al. (2018) captures price-demand dependence using cross moments under random market prices.

3.2. Supply

3.2.1. Capacity. Although uncertain production capacity under normal operating conditions can be easily incorporated into many robust OM models, it seems this direction has not been pursued in literature. This is probably because in a production system under normal operating conditions, capacity information is typically well known and has relatively low uncertainty. Uncertain capacity has primarily been investigated in the context of supply risk (i.e., capacity loss).

3.2.2. Lead Time. Inventory management with random lead time is known to be challenging. Robust OM provides an effective means to model lead time uncertainty. Movahed and Zhang (2015) considered uncertain lead times with finite scenarios. Thorsen and Yao (2017) considered uncertain sets motivated by the central limit theorem.

3.2.3. Capacity Loss. In robust OM, uncertain capacity loss has been investigated primarily in the special case of supply disruptions, that is, complete capacity loss. An et al. (2014) studied the robust p-median facility location problem where a pre-specified number of facilities can be disrupted. Lu et al. (2015) and Zhao and Freeman (2018) considered ambiguous disruption distributions with known marginals. Zhao and Freeman (2016) considered multiple levels of capacity loss (as well as random yield) with ambiguous distributions. Shen et al. (2017) studied network design problems with uncertain arc capacities.

3.2.4. Yield and Quality. Both random yield and random capacity have been extensively studied in the OM literature. The high level of yield uncertainty in certain industries, such as, semiconductors, lumber, and petroleum, motivated the application of robust optimization to production planning in these industries (Ng et al. 2010, Zanjani et al. 2010). Ng et al. (2012) considered a special case of yield uncertainty in co-production technology, where a number of different products are generated from the same production process. In a supply chain setting, Park and Lee (2016) examined single period inventory control from multiple unreliable suppliers with ambiguous yield distributions.

3.2.5. Remanufacturing and Consumer Returns. Closed-loop supply chain management is a relatively new research area, where reverse material flows, such as remanufacturing and consumer returns, occur because of quality problems or customer dissatisfaction. Although we categorize them under uncertainty related supply, remanufacturing and consumer returns are related to both supply and demand. For example, remanufacturing may provide recyclable materials but may also require additional work. Whereas some of the returned product can be resold directly, the amount of consumer returns usually depends on the amount of sales in the previous period. The high level of uncertainty and the possible lack of data in remanufacturing and consumer returns make robust OM a promising modeling framework for dealing with these issues. Keyvanshokooh et al. (2016) and Wei et al. (2011) considered uncertain returns represented by scenarios and intervals, respectively. In addition to return volume, uncertainty in the quality of returned/recycled products has also been examined (Jiao et al. 2018).

3.3. Time in Scheduling and Routing

In the previous subsections, we have discussed uncertain interarrival and service times and uncertain lead time, which may arise in queueing systems and inventory systems. Dealing with uncertain time durations is a central research topic in other OM areas, such as production scheduling, project management, and vehicle routing.

3.3.1. Job/Task Completion Time. In production scheduling, Daniels and Kouvelis (1995) considered uncertainty processing times, where uncertainty is represented by finite scenarios. Kasperski (2005b) considered the case of interval uncertainty. Chang et al. (2017) considered distribution ambiguity where only the mean and covariance matrix are known. Like uncertain processing times in production scheduling, uncertain task completion times in project management have also motivated the use of robust optimization. Cohen et al. (2007) considered interval and ellipsoidal uncertainty sets for project crashing problems with uncertain task durations. Chen et al. (2007, 2008) and Goh and Hall (2013) considered different types of distribution ambiguity for task durations. Alternatively, Wiesemann et al. (2012a, b) considered uncertainty sets for the work content, which enables a more general model for project crashing with resource allocation.

3.3.2. Travel Time. Uncertainty in travel time (or distance) has been considered in various routing problems, such as the shortest path problem (Yu and Yang 1998, Zhang et al. 2018b), the traveling salesman problem (Montemanni et al. 2007), and the vehicle routing problem (Sungur et al. 2008). Most of the literature on robust routing assumes finite scenarios or interval uncertainty. Jaillet et al. (2016) was the first to consider different types of distribution ambiguity.

3.4. Monetary Terms

3.4.1. Cost. Classical OM research usually assumes cost is deterministic. However, cost uncertainty has been considered by many in robust OM. Different types of cost considered include fixed ordering and purchase cost (Yu 1997), production and labor-related cost (Leung and Wu 2004), fixed location and transportation cost (Kouvelis and Yu 1997), and capacity expansion cost (Marin and Salmeron 2001).

3.4.2. Price. Price uncertainty has been widely studied in power systems using robust optimization, because the market price for electricity is typically uncertain at the time of planning (Fan et al. 2014). In other OM areas, price is usually either assumed to be deterministic or considered a decision. Fu et al. (2018) pointed out that market price in OM is usually highly volatile. They developed a distributionally robust optimization model for supply chain contracting, in which both demand and price have ambiguous distributions.

3.5. Preference and Utility

Most of OM models are concerned with decision-making, for which a criterion must be specified. Furthermore, many OM problems require modeling the behavior of consumers or agents based on their own utility or preference. In traditional models, both the principal decision maker's criterion and how the agents behave are precisely known. However, both types of knowledge may be unavailable in practice, which has motivated the application of robust optimization.

3.5.1. Valuation. Wang et al. (2014) and Caldentey et al. (2017) examined pricing when the consumer's willingness-to-pay (valuation) is only known to lie within an interval. Pinar and Kizilkale (2017) investigated optimal screening mechanisms when selling to a buyer with ambiguous valuation distributions on a finite support. He et al. (2017) considered a satisficing model for availability-driven service adoption, where the users have uncertain utilities for service availability at different locations and the utilities have an ambiguous distribution defined by the first two moments.

3.5.2. Utility Function and Risk Measure. Preferences over random outcomes or multiple objectives are usually modeled with utility functions or risk measures. In most of the classical OM literature, utility functions and risk measures are assumed to be given. However, it is also common that decision makers lack a precise knowledge of the utility function or risk measures. Recently, robust optimization has been used to tackle such ambiguity (Armbruster and Delage 2015, Haskell et al. 2016, Hu and Mehrotra 2012). This stream of research is closely related to stochastic dominance, which we discuss in more detail in section 5.3.

3.5.3. Preference and Choice. When multiple alternatives are available, different consumer valuations of the alternatives will result in different preferences and choices. Various robust models have been proposed for the modeling of preference and choice uncertainty. Natarajan et al. (2009b) considered ambiguous utility distributions defined by marginal distributions and marginal moments, and they found the resulting choice model to be more general than the classical multinomial logit model, which requires the independence of irrelevant alternatives (IIA) assumption. Rusmevichientong and Topaloglu (2012) considered parameter uncertainty in the multinomial logit model. Farias et al. (2013) considered nonparametric uncertainty sets consisting of all choice models that are consistent with observed data. Bertsimas and O'Hair (2013) considered uncertain coefficients in linear utility functions for multiple features, and they constructed uncertainty sets for the coefficients consistent with questionnaire responses.

4. Successes of Robust Operations Management

As we have noted in the previous sections, robust optimization frameworks with diverse forms of uncertainty and decisions criteria have been applied to a full spectrum of OM problems. There are a number of reasons for the tremendous success of robust OM.

First, robust OM addresses model uncertainties and ambiguities that are prevalent in practice. The use of robust optimization not only provides an alternative way to represent uncertainty, but has also broadened the scope of uncertainty modeling in OM. New sources of uncertainty, such as ambiguous distribution, dependence, utility, and risk preferences, have greatly enriched the OM literature.

Second, robust OM can be equipped with a variety of decision criteria, which may better reflect practical business goals and decision-making behaviors. Worst-case objectives and robust feasibility are well suited for applications in risk management, emergency planning, and public services. Regret minimization and satisficing-inspired models capture theoretical and behavior issues that are not addressed in traditional models.

Third, despite their versatile forms and ample features, robust OM models have generated new structural results and valuable insights, from the pioneering work of Scarf (1957) and new closed-form results in inventory management (e.g., Mamani et al. 2017, Wagner 2018) to optimal time allowances in appointment scheduling (e.g., Mak et al. 2015) and long chain sparse structure in process flexibility (e.g., Wang and Zhang 2015).

Last but not least, through their compact reformulations, robust OM models have empowered us to analyze uncertain problems without increasing the computational burden rendered by their deterministic counterparts. Special structures of the worst-case distributions in distributionally robust models have simplified the nominal stochastic optimization problems and enabled the integration of more random factors into deterministic optimization models.

5. Research Opportunities and Challenges

5.1. Robust Operations Management and Data Science

Recently, fueled by the explosive growth of available business data and research advancement in computer science and statistics, there has been unprecedented interest in applying data science, big data analytics, and machine learning to OM (Choi et al. 2017, Corbett 2017, Feng and Shanthikumar 2018a, Fisher and Raman 2018, Simchi-Levi 2013, Swaminathan 2018). The big data era has also brought forth research opportunities for robust OM.

5.1.1. The Value of Robustness in the Big Data Era. Some may argue that robust optimization is primarily motivated by a data-scarce environment to address uncertainty and ambiguity. However, as we observed in sections 2 and 3, robust optimization has proven to be extremely useful in modeling diverse decision criteria, relaxing restrictive structural assumptions such as IIA, and assisting multistage decision-making. In many cases, it also results in tractable reformulations and closed-form solutions, enabling efficient computation and providing unique insights. These benefits will not be diminished as more data become available. The optimization framework of robust OM facilitates its integration with big data analytics, such as machine learning (e.g., Kim and Lim 2015). Furthermore, robust optimization has been shown to have deep theoretical connections to statistical learning (Bertsimas and Copenhaver 2018, Rahimian and Mehrotra 2019, Xu et al. 2009).

5.1.2. Using Big Data in Robust Operations Management. As we mentioned in the previous subsection, robust OM will contribute to research initiatives that integrate OM and data science. Data can also be used effectively in robust OM. There has been significant progress in the understanding of how to construct uncertainty sets from data for different types of robust optimization models (Bertsimas and Thiele 2006a, Bertsimas et al. 2018a, Delage and Ye 2010, Esfahani and Kuhn 2017, Jiang and Guan 2016). Hong et al. (2017) proposed a data-splitting procedure with initial construction followed by calibration to provide nonparametric probabilistic guarantee. In OM, it is common to have censored data. Robust optimization can be applied to utilize censored data more effectively. Big data can also expand the sources of uncertainty for robust OM. In robust optimization, uncertain parameters are usually assumed to be linearly dependent on a number of primitive factors. Big data with a large number of features (factors) can be used to improve OM decisions and deliver personalized products or services (Ban and Keskin 2017, Ban and Rudin 2019, Feng and Shanthikumar 2018a). Robust optimization based on a large number of features provides an effective means of achieving these goals.

Constructing uncertainty sets from data and representing uncertainty with features are examples of *data-driven* robust optimization, in which the output of data analytics (e.g., statistics) is used as an input for decision-making. In a *data-integrated* approach, data are fully integrated into the decision-making process. One example of data-integrated approaches is the *operational statistics* or *operational learning* (Chu et al. 2008, Liyanage and Shanthikumar 2005), which finds the optimal decision rule for a predetermined distribution. It will be interesting to develop this approach in the context of distributional ambiguity.

5.2. Robust Operations Management in Emerging Areas

Recent years have witnessed the emergence of numerous innovative business models enabled by new technology or operating systems, as well as OM models with nontraditional features motivated by increased social-environmental awareness (Tang 2015). Robust optimization will prove useful in modeling and improving the operations in these emerging areas in multiple ways.

First, operations in emerging areas may have an intrinsically high level of uncertainty and ambiguity, which may arise from system dynamics, technology advancement, user behavior, market response, etc. Robust optimization can provide flexible representation of these uncertainties. For example, Lyu and Teo (2019) studied the design of locker networks for efficient last-mile delivery. Robust optimization is applied to address the uncertainty in users' residential and work address relationships. He et al. (2017) investigated carsharing service system planning under significant uncertainty in user adoption behavior. They assumed users gain utility from service coverage at different locations and adopted the moment model to address distributional ambiguity of user utility.

Second, OM models for emerging areas may have nontraditional objectives or constraints. For example, research has shown that about 70% of technology startup companies fail within 20 months after first raising funding (CB Insights 2018). Under the high pressure of survival, a traditional profit-maximization objective may not be suitable for modeling these types of operations. Robust decision criteria such as optimizing the worst case, goal-driven satisficing, *p*-robustness, etc., will be better aligned with practical concerns. For example, Mak et al. (2013) examined infrastructure planning of battery swapping services for electric vehicles. They adopted a robust goal-driven model to maximize the probability of sustaining business. The fact that the first company to provide battery swapping services, Better Place, went bankrupt in May 2013, only one year after launching the service (Fortune 2013), shows the need for applying robust goal-driven models in practice.

Third, because many OM problems in emerging areas involve new features that must be incorporated into traditional models, the study of these problems often requires integrated modeling (Mak and Shen 2016). One example is the operation of "smart cities." As noted by Qi and Shen (2018), OM problems for smart-city often require the integration of multiple systems that are traditionally studied separately, such as energy systems, transportation systems, and various urban service systems. Robust optimization provides flexible and tractable formulations that facilitate model integration for such problems. Certain robust OM models also provide closed-form expressions that can serve as building blocks for integrated models with larger scopes.

5.3. Robust Operations Management and Stochastic Orders

The use of risk measures and stochastic orders is common in OM literature. It has been shown that there exist profound and elegant correspondences between distributionally robust optimization and risk measures (e.g., Ben-Tal and Teboulle 2007, , Ben-Tal et al. 2010, Bertsimas and Brown 2009, Gotoh et al. 2018, Natarajan et al. 2009a). The consistency between risk measures and stochastic orders has also been widely studied (e.g., Bäuerle and Müller 2006). However, the interplay between robust optimization and stochastic orders has not received much attention until recently.

5.3.1. Robustness from Stochastic Orders. In some cases, certain stochastic orders naturally lead to worst-case distributions in robust optimization. One such example is the *supermodular order* (Shaked and Shanthikumar 1997). Recall the marginal distribution model discussed in section 2.1, where the expected objective value is optimized within a Frèchet class. If the objective function is supermodular in the random parameters, which is common in OM models such as inventory control (Mak and Shen 2014) and facility location (Lu et al. 2015), then the supermodular order leads directly to the worst-case distribution, known as the Frèchet upper bound (Shaked and Shanthikumar 2007). To take another example, Feng and Shanthikumar (2018b) considered the classical pricing problem where demand is determined by the price and the buyer's random valuation. They defined the *scaled pricing stochastic order* based on the comparison of optimal expected revenues resulting from different consumer valuation distributions. Hence, for a certain distribution family, the distribution that is stochastically smaller than all other distributions in the scaled pricing order is the worst-case distribution for the distributionally robust pricing problem on that distribution family.

Uncovering more connections between distributionally robust optimization and stochastic orders for specific robust OM problems will be an interesting research direction. It is also important to investigate how results from stochastic ordering can be combined with other commonly used forms of distributional information such as moments, dispersion, and divergence. For example, Mak and Shen (2014) showed that for a distributionally robust optimization problem with a supermodular objective function, when the marginal distribution is not known, but only the marginal moments are known, the worst-case distribution is still the upper-bound of a Fréchet class with the corresponding marginal moments.

5.3.2. Robust Stochastic Dominance. Stochastic optimization with stochastic dominance constraints has been widely applied to portfolio management in finance. In OM, it has been applied to supply chain management (Escudero et al. 2015) and transportation (Nie et al. 2012). Stochastic dominance is closely related to robust optimization. First, many stochastic dominance or stochastic ordering constraints, in their nominal (i.e., "nonrobust") form, can be readily viewed as robust constraints. For example, second-order stochastic dominance can be viewed as a robust constraint that must hold for all increasing concave functions. This is especially relevant in robust OM, because it guarantees the preference of the random outcome over some given benchmark under all increasing concave utility functions, and hence provides robustness with respect to ambiguity in utility.

Second, one can further define robust stochastic dominance using distributionally robust optimization. As noted by Haskell et al. (2016), this captures ambiguity in both the utility function and the distribution. Dentcheva and Ruszczyński (2010) considered robust second-order stochastic dominance, where dominance must hold under any probability measure within a subset. Jiang and Guan (2015) considered robust first-order stochastic dominance constraints that are defined differently from those discussed in Dentcheva and Ruszczyński (2010). It is important to further explore the use of stochastic dominance and other stochastic ordering constraints to represent ambiguity in utility and to enable adjusting the level of conservativeness. Incorporating these constraints to specific robust OM problems may lead to novel and interesting insights. Stochastic optimization problems with stochastic dominance constraints are generally hard to solve. Practical optimization methods for these problems have recently gained increased attention (see Haskell et al. 2013, 2017a,b, and the references therein). Solving large-scale robust OM problems with stochastic dominance constraints remains a challenging problem.

5.3.3. Ambiguity Sets with Stochastic Ordering. The stochastic dominance and robust stochastic dominance constraints discussed in the previous subsection are imposed upon the decisions. It is also possible to use stochastic dominance to refine the uncertainty set for distributionally robust optimization. Chen et al. (2019) proposed uncertainty sets with second-order stochastic dominance as a special case of their general framework of infinitely constrained uncertainty sets.

In general, stochastic ordering can be used to refine commonly used uncertainty sets to incorporate more domain knowledge and avoid being overly pessimistic. This approach may be of particular interest to robust OM. For example, in joint pricing-inventory management where the demand function and demand distribution are not precisely known, one could employ stochastic orders to model how the demand magnitude and variability change with respect to the price. To take another example, in supply chain sourcing from multiple unreliable suppliers, it is typically assumed that the suppliers can be ordered in terms of their costs or reliability. Most existing studies have either assumed no ambiguity or resorted to parametric representations, which may not be available in practice. Stochastic orders can be adopted to incorporate the knowledge that the suppliers are known to be ordered in a certain way, but no parametric representations are available. As noted by Chen et al. (2019), distributionally robust optimization problems with stochastic ordering conditions in the ambiguity sets may not have tractable reformulations. Designing tractable approximation methods for these problems is key to their applications in OM.

5.4. Efficient Computation for Robust Operations Management

Computation has always been an important aspect of robust optimization. As Bertsimas et al. (2011) pointed out, the "massive flurry of interest" in robust optimization was sparked in the late 1990s by tractable reformulations of different robust counterparts, as well as by significant advances in computation and solution methods for convex optimization. In contrast to models that seek to provide structural results, many robust OM models are computation-driven; that is, they seek to provide implementable numerical solutions to problems of practical sizes. Recently, many of such models, especially those with distribution ambiguity defined by moments, have been reformulated as problems that are NP-hard in general but have tractable approximations or efficient computational methods, such as copositive programming (e.g., Gao et al. 2019, Kong et al. 2013, 2020, Yan et al. 2017) and mixed-integer second-order cone programming (e.g., Mak et al. 2013, 2015). Such reformulations enable researchers to leverage developments in solving copositive programming and mixed-integer conic programming to facilitate robust OM computation.

As we mentioned in section 2.4, robust optimization problems can be solved using either the compact reformulation approach or the cutting plane approach. Recently, the latter has been shown to have better performance for certain problems (Bertsimas et al. 2016, Fischetti and Monaci 2012). Decomposition methods based on cutting planes have also been developed for two-stage robust optimization problems with very promising computational performance (e.g., Zeng and Zhao 2013).

6. Concluding Remarks

Robust optimization has been applied to a full spectrum of OM problems in order to incorporate various sources of uncertainty and ambiguity with flexible modeling frameworks. Robust OM research has enjoyed tremendous success in addressing model uncertainty, extending the scope of uncertainty modeling, providing alternative practical decision criteria, generating structural results and novel insights, and improving tractability and computation efficiency. Robust OM will both contribute to and benefit from the integration of OM with data science. It has unique advantages in modeling, analyzing, and optimizing operations in emerging areas. New theory and applications can stem from the interplay between robustness and stochastic orders, and solutions can be facilitated by leveraging advances in conic optimization and developing new decomposition algorithms.

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References

Adida, E., G. Perakis. 2006. A robust optimization approach to dynamic pricing and inventory control with no backorders. *Math. Program.* **107**(1-2): 97–129.

Adida, E., G. Perakis. 2010. Dynamic pricing and inventory control: Robust vs. stochastic uncertainty models-a computational study. *Ann. Oper. Res.* **181**(1): 125–157.

Agrawal, S., Y. Ding, A. Saberi, Y. Ye. 2012. Price of correlations in stochastic optimization. *Oper. Res.* **60**(1): 150–162.

An, Y., B. Zeng, Y. Zhang, L. Zhao. 2014. Reliable p-median facility location problem: Two-stage robust models and algorithms. *Trans Res B: Methodol.* **64**: 54–72.

Ardestani-Jaafari, A., E. Delage. 2016. Robust optimization of sums of piecewise linear functions with application to inventory problems. *Oper. Res.* **64**(2): 474–494.

Armbruster, B., E. Delage. 2015. Decision making under uncertainty when preference information is incomplete. *Management Sci.* **61**(1): 111–128.

Atamtürk, A., M. Zhang. 2007. Two-stage robust network flow and design under demand uncertainty. *Oper. Res.* **55**(4): 662–673.

Averbakh, I., O. Berman. 1997. Minimax regret p-center location on a network with demand uncertainty. *Location Sci.* **5**(4): 247–254.

Averbakh, I., O. Berman. 2000a. Algorithms for the robust 1-center problem on a tree. *Eur. J. Oper. Res.* **123**(2): 292–302.

Averbakh, I., O. Berman. 2000b. Minmax regret median location on a network under uncertainty. *INFORMS J. Comput.* **12**(2): 104–110.

Averbakh, I., V. Lebedev. 2005. On the complexity of minmax regret linear programming. *Eur. J. Oper. Res.* **160**(1): 227–231.

Ban, G. Y., N. B. Keskin. 2017. Personalized dynamic pricing with machine learning. SSRN Scholarly Paper ID 2972985, Social Science Research Network, Rochester, NY

Ban, G. Y., C. Rudin. 2019. The big data newsvendor: Practical insights from machine learning. *Oper. Res.* **67**(1): 90–108.

Bandi, C., D. Bertsimas. 2012. Tractable stochastic analysis in high dimensions via robust optimization. *Math. Program.* **134**(1): 23–70.

Bandi, C., D. Bertsimas, N. Youssef. 2015. Robust queueing theory. *Oper. Res.* **63**(3): 676–700.

Baron, O., J. Milner, H. Naseraldin. 2011. Facility location: A robust optimization approach. *Prod. Oper. Manag.* **20**(5): 772–785.

Basciftci, B., S. Ahmed, S. Shen. (2019) Distributionally robust facility location problem under decision-dependent stochastic demand. arXiv preprint arXiv:1912.05577.

Bäuerle N., A. Müller. 2006. Stochastic orders and risk measures: Consistency and bounds. *Insur. Math. Econ.* **38**(1): 132–148.

Ben-Tal, A., A. Nemirovski. 1999. Robust solutions of uncertain linear programs. *Oper. Res. Lett.* **25**(1): 1–13.

Ben-Tal, A., M. Teboulle. 2007. An old-new concept of convex risk measures: The optimized certainty equivalent. *Math. Financ.* **17**(3): 449–476.

Ben-Tal, A., A. Goryashko, E. Guslitzer, A. Nemirovski. 2004. Adjustable robust solutions of uncertain linear programs. *Math. Program.* **99**(2): 351–376.

Ben-Tal, A., B. Golany, A. Nemirovski, J. P. Vial. 2005. Retailer-supplier flexible commitments contracts: A robust optimization approach. *Manuf. Serv. Oper. Manag.* **7**(3): 248–271.

Ben-Tal, A., L. E. Ghaoui, A. Nemirovski. 2009a. *Robust Optimization*. Princeton University Press, Princeton.

Ben-Tal, A., B. Golany, S. Shtern. 2009b. Robust multi-echelon multi-period inventory control. *Eur. J. Oper. Res.* **199**(3): 922–935.

Ben-Tal, A., D. Bertsimas, D. B. Brown. 2010. A soft robust model for optimization under ambiguity. *Oper. Res.* **58**(4): 1220–1234.

Ben-Tal, A., B. Do Chung, S. R. Mandala, T. Yao. 2011. Robust optimization for emergency logistics planning: Risk mitigation in humanitarian relief supply chains. *Trans Res B: Methodol.* **45**(8): 1177–1189.

Ben-Tal, A., D. Den Hertog, A. DeWaegenaere, B. Melenberg, G. Rennen. 2013. Robust solutions of optimization problems affected by uncertain probabilities. *Management Sci.* **59**(2): 341–357.

Berman, O., I. Hajizadeh, D. Krass. 2013. The maximum covering problem with travel time uncertainty. *IIE Trans.* **45**(1): 81–96.

Bertsimas, D., D. B. Brown. 2009. Constructing uncertainty sets for robust linear optimization. *Oper. Res.* **57**(6): 1483–1495.

Bertsimas, D., M. S. Copenhaver. 2018. Characterization of the equivalence of robustification and regularization in linear and matrix regression. *Eur. J. Oper. Res.* **270**(3): 931–942.

Bertsimas, D., A. O'Hair. 2013. Learning preferences under noise and loss aversion: An optimization approach. *Oper. Res.* **61**(5): 1190–1199.

Bertsimas, D., M. Sim. 2003. Robust discrete optimization and network flows. *Math. Program.* **98**(1–3): 49–71.

Bertsimas, D., M. Sim. 2004. The price of robustness. *Oper. Res.* **52**(1): 35–53.

Bertsimas, D., A. Thiele. 2006a. Robust and data-driven optimization: Modern decision-making under uncertainty. *INFORMS Tutorials in Operations Research: Models, Methods, and Applications for Innovative Decision Making*, 95–122.

Bertsimas, D., A. Thiele. 2006b. A robust optimization approach to inventory theory. *Oper. Res.* **54**(1): 150–168.

Bertsimas, D., K. Natarajan, C. P. Teo. 2006. Persistence in discrete optimization under data uncertainty. *Math. Program.* **108**(2–3): 251–274.

Bertsimas, D., D. Brown, C. Caramanis. 2011. Theory and applications of robust optimization. *SIAM Rev.* **53**(3): 464–501.

Bertsimas, D., V. Goyal, B. Y. Lu. 2015. A tight characterization of the performance of static solutions in two-stage adjustable robust linear optimization. *Math. Program.* **150**(2): 281–319.

Bertsimas, D., I. Dunning, M. Lubin. 2016. Reformulation vs. cutting-planes for robust optimization. *CMS* **13**(2): 195–217.

Bertsimas, D., V. Gupta, N. Kallus. 2018a. Data-driven robust optimization. *Math. Program.* **167**(2): 235–292.

Bertsimas, D., M. Sim, M. Zhang. (2018b) Adaptive distributionally robust optimization. *Management Sci.* **65**(2): 604–618.

Bienstock, D., N. Ozbay. 2008. Computing robust basestock levels. *Discrete Optim.* **5**(2): 389–414.

Bruni, M. E., Pugliese L. Di Puglia, P. Beraldi, F. Guerriero. 2017. An adjustable robust optimization model for the resource-constrained project scheduling problem with uncertain activity durations. *Omega*. **71**: 66–84.

Caldentey, R., Y. Liu, I. Lobel. 2017. Intertemporal pricing under minimax regret. *Oper. Res.* **65**(1): 104–129.

CB Insights 2018. 253 Startup Failure Post-Mortems. Available at https://www.cbinsights.com/research/startup-failure-post-mortem/ (accessed date July 1, 2018).

Chan, T. C. Y., Z. J. M. Shen, A. Siddiq. 2017. Robust defibrillator deployment under cardiac arrest location uncertainty via row-and-column generation. *Oper. Res.* **66**(2): 358–379.

Chang, Z., S. Song, Y. Zhang, J. Y. Ding, R. Zhang, R. Chiong. 2017. Distributionally robust single machine scheduling with risk aversion. *Eur. J. Oper. Res.* **256**(1): 261–274.

Chen, W., M. Sim. 2009. Goal-driven optimization. *Oper. Res.* **57**(2): 342–357.

Chen, X., Y. Zhang. 2009. Uncertain linear programs: Extended affinely adjustable robust counterparts. *Oper. Res.* **57**(6): 1469–1482.

Chen, G., M. S. Daskin, Z. J. M. Shen, S. Uryasev. 2006. The α-reliable mean-excess regret model for stochastic facility location modeling. *Nav. Res. Logist.* **53**(7): 617–626.

Chen, X., M. Sim, P. Sun. 2007. A robust optimization perspective on stochastic programming. *Oper. Res.* **55**(6): 1058–1071.

Chen, X., M. Sim, P. Sun, J. Zhang. 2008. A linear decision-based approximation approach to stochastic programming. *Oper. Res.* **56**(2): 344–357.

Chen, Z. L., N. G. Hall, H. Kellerer. 2017. Dynamic pricing to minimize maximum regret. *Prod. Oper. Manag.* **26**(1): 47–63.

Chen, L., W. Ma, K. Natarajan, D, Simchi-Levi, Z, Yan. 2018a. Distributionally robust linear and discrete optimization with marginals. SSRN Scholarly Paper ID 3159473, Social Science Research Network, Rochester, NY.

Chen, Z., M. Sim, P. Xiong. 2018b. Adaptive robust optimization with scenario-wise ambiguity sets. Available at Optimization Online.

Chen, Z., M. Sim, H. Xu. 2019. Distributionally robust optimization with infinitely constrained ambiguity sets. *Oper. Res.* **67**(5): 1328–1344.

Choi, T. M., S. W. Wallace, Y. Wang. 2017. Big data analytics in operations management. *Prod. Oper. Manag.* **27**(10): 1868–1883

Chu, L. Y., J. G. Shanthikumar, Z. J. M. Shen. 2008. Solving operational statistics via a Bayesian analysis. *Oper. Res. Lett.* **36**(1): 110–116.

Cohen, I., B. Golany, A, Shtub (2007) The stochastic time–cost tradeoff problem: A robust optimization approach. *Netw. Int. J.* **49**(2): 175–188.

Corbett, C. J. 2017. How sustainable is big data? *Prod. Oper. Manag.* **27**(9): 1685–1695.

Daniels, R. L., P. Kouvelis. 1995. Robust scheduling to hedge against processing time uncertainty in single-stage production. *Management Sci.* **41**(2): 363–376.

Daskin, M. S., S. M. Hesse, C. S. Revelle. 1997. α-Reliable *p*-minimax regret: A new model for strategic facility location modeling. *Location Sci.* **5**(4): 227–246.

Delage, E., Y. Ye. 2010. Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Oper. Res.* **58**(3): 595–612.

Deng, Y., S. Shen, B. Denton. 2019. Chance-constrained surgery planning under conditions of limited and ambiguous data. *INFORMS J. Comput.* **31**(3): 559–575.

Dentcheva, D., Ruszczyński A, 2010. Robust stochastic dominance and its application to risk-averse optimization. *Math. Program.* **123**(1): 85—100.

Drwal, M., R. Rischke. 2016. Complexity of interval minmax regret scheduling on parallel identical machines with total completion time criterion. *Oper. Res. Lett.* **44**(3): 354–358.

El-Amine, H., E. K. Bish, D. R. Bish. 2018. Robust postdonation blood screening under prevalence rate uncertainty. *Oper. Res.* **66**(1): 1–17.

Erdoğan, E., G. Iyengar. 2006. Ambiguous chance constrained problems and robust optimization. *Math. Program.* **107**(1–2): 37–61.

Escudero, L. F., J. F. Monge, D. Romero Morales. 2015. An SDP approach for multiperiod mixed 0–1 linear programming models with stochastic dominance constraints for risk management. *Comput. Oper. Res.* **58**: 32–40.

Esfahani, P. M., D. Kuhn. 2017. Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Math. Program.* **171**: 115–166.

Fan, L., J. Wang, R. Jiang, Y. Guan. 2014. Min-Max regret bidding strategy for thermal generator considering price uncertainty. *IEEE Trans. Power Syst.* **29**(5): 2169–2179.

Farias, V. F., S. Jagabathula, D. Shah. 2013. A nonparametric approach to modeling choice with limited data. *Management Sci.* **59**(2): 305–322.

Feng, Q., J. G. Shanthikumar. 2018a. How research in production and operations management may evolve in the era of big data. *Prod. Oper. Manag.* **27**(9): 1670–1684.

Feng, Q., J. G. Shanthikumar. 2018b. Posted pricing vs. bargaining in sequential selling process. *Oper. Res.* **66**(1): 92–103.

Fischetti, M., M. Monaci. 2009. *Light robustness. Robust and Online Large-Scale Optimization*, Springer, Berlin, 61–84.

Fischetti, M., M. Monaci. 2012. Cutting plane vs. compact formulations for uncertain (integer) linear programs. *Math. Program. Comput.* **4**(3): 239–273.

Fisher, M., A. Raman. 2018. Using data and big data in retailing. *Prod. Oper. Manag.* **27**(9): 1665–1669.

Fortune. 2013. Exclusive: Better Place to File for Bankruptcy. Available at http://fortune.com/2013/05/24/exclusive-better-place-to-file-for-bankruptcy/ (accessed date July 1, 2018).

Fu, Q., C. K. Sim, C. P. Teo. 2018. Profit sharing agreements in decentralized supply chains: A distributionally robust approach. *Oper. Res.* **66**(2): 500–513.

Gabrel, V., C. Murat, A. Thiele. 2014. Recent advances in robust optimization: An overview. *Eur. J. Oper. Res.* **235**(3): 471–483.

Gallego, G., I. Moon. 1993. The distribution free newsboy problem: Review and extensions. *J. Oper. Res. Soc.* **44**(8): 825–834.

Gao, S. Y., D. Simchi-Levi, C. P. Teo, Z. Yan. 2019. Disruption risk mitigation in supply chains: The risk exposure index revisited. *Oper. Res.* **67**(3): 831–852.

Goh, J., N. G. Hall. 2013. Total cost control in project management via satisficing. *Management Sci.* **59**(6): 1354–1372.

Goh, J., M. Sim. 2010. Distributionally robust optimization and its tractable approximations. *Oper. Res.* **58**(4-part–1): 902–917.

Goh, J., M. Sim. 2011. Robust optimization made easy with ROME. *Oper. Res.* **59**(4): 973–985.

Gorissen, B. L., I. Yanıkoğlu, D. den Hertog. 2015. A practical guide to robust optimization. *Omega* **53**: 124–137.

Gotoh J. Y., M. J. Kim, A. E. B. Lim. 2018. Robust empirical optimization is almost the same as mean-variance optimization. *Oper. Res. Lett.* **46**(4): 448–452.

Gutierrez, G. J., P. Kouvelis, A. A. Kurawarwala. 1996. A robustness approach to uncapacitated network design problems. *Eur. J. Oper. Res.* **94**(2): 362–376.

Hanasusanto, G. A., D. Kuhn, S. W. Wallace, S. Zymler. 2015. Distributionally robust multi-item newsvendor problems with multimodal demand distributions. *Math. Program.* **152**(1–2): 1–32.

Hao, Z., L. He, Z. Hu, J. Jiang. 2019. Robust vehicle pre-allocation with uncertain covariates. *Prod. Oper. Manag.* **29**: 955–972

Haskell, W. B., J. G. Shanthikumar, Z. M. Shen. 2013. Optimization with a class of multivariate integral stochastic order constraints. *Ann. Oper. Res.* **206**(1): 147–162.

Haskell, W. B., L. Fu, M. Dessouky. 2016. Ambiguity in risk preferences in robust stochastic optimization. *Eur. J. Oper. Res.* **254**(1): 214–225.

Haskell, W., J. Shanthikumar, Z. Shen. 2017a. Primal-dual algorithms for optimization with stochastic dominance. *SIAM J. Optim.* **27**(1): 34–66.

Haskell W. B., J. G. Shanthikumar, Z. M. Shen. 2017b. Aspects of optimization with stochastic dominance. *Ann. Oper. Res.* **253**(1): 247–273.

Haviv, M., R. S. Randhawa. 2014. Pricing in queues without demand information. *Manuf. Serv. Oper. Manag.* **16**(3): 401–411.

He, L., H. Y. Mak, Y. Rong, Z. J. M. Shen. 2017. Service region design for urban electric vehicle sharing systems. *Manuf. Serv. Oper. Manag.* **19**(2): 309–327.

He, L., Z. Hu, M. Zhang. 2020. Robust repositioning for vehicle sharing. *Manuf. Serv. Oper. Manag.* **22**(2): 241–256.

Hong, L. J., Z. Huang, H. Lam. 2017. Learning-based Robust Optimization: Procedures and Statistical Guarantees. arXiv:1704.04342 [math].

Hu, Z., L. J. Hong. 2013. Kullback-Leibler divergence constrained distributionally robust optimization. Available at Optimization Online.

Hu, J., S. Mehrotra. 2012. Robust and stochastically weighted multiobjective optimization models and reformulations. *Oper. Res.* **60**(4): 936–953.

Iancu, D. A., N. Trichakis. 2014. Pareto efficiency in robust optimization. *Management Sci* **60**(1): 130–147.

Jaillet, P., J. Qi, M. Sim. 2016. Routing optimization under uncertainty. *Oper. Res.* **64**(1): 186–200.

Jain, A., A. E. B. Lim, J. G. Shanthikumar. 2010. On the optimality of threshold control in queues with model uncertainty. *Queueing Syst.* **65**(2): 157–174.

Jiang, R., Y. Guan. 2015. Risk-averse two-stage stochastic program with distributional ambiguity. Available on Optimization Online.

Jiang, R., Y. Guan. 2016. Data-driven chance constrained stochastic program. *Math. Program.* **158**(1–2): 291–327.

Jiang, R., J. Wang, Y. Guan. 2012. Robust unit commitment with wind power and pumped storage hydro. *IEEE Trans. Power Syst.* **27**(2): 800.

Jiang, R., J. Wang, M. Zhang, Y. Guan. 2013. Two-stage minimax regret robust unit commitment. *IEEE Trans. Power Syst.* **28**(3): 2271–2282.

Jiao, Z., L. Ran, Y. Zhang, Z. Li, W. Zhang. 2018. Data-driven approaches to integrated closed-loop sustainable supply chain design under multi-uncertainties. *J. Clean. Prod.* **185**: 105–127.

Kalaï, R., C. Lamboray, D. Vanderpooten. 2012. Lexicographic α-robustness: An alternative to min-max criteria. *Eur. J. Oper. Res.* **220**(3): 722–728.

Kasperski, A. 2005a. Minimizing maximal regret in the single machine sequencing problem with maximum lateness criterion. *Oper. Res. Lett.* **33**(4): 431–436.

Kasperski, A. 2005b. Minimizing maximal regret in the single machine sequencing problem with maximum lateness criterion. *Oper. Res. Lett.* **33**(4): 431–436.

Kasugai, H., T. Kasegai. 1960. Characteristics of dynamic maximin ordering policy. *J. Oper. Res. Soc. Jpn.* **3**(1): 11–26.

Kasugai, H., T. Kasegai. 1961. Note on minimax regret ordering policy-static and dynamic solutions and a comparison to maximin policy. *J. Oper. Res. Soc. Jpn.* **3**(4): 155–169.

Keyvanshokooh, E., S. M. Ryan, E. Kabir. 2016. Hybrid robust and stochastic optimization for closed-loop supply chain network design using accelerated Benders decomposition. *Eur. J. Oper. Res.* **249**(1): 76–92.

Kim, M. J., A. E. Lim. 2015. Robust multiarmed bandit problems. *Management Sci.* **62**(1): 264–285.

Klabjan, D., D. Simchi-Levi, M. Song. 2013. Robust stochastic lot-sizing by means of histograms. *Prod. Oper. Manag.* **22**(3): 691–710.

Kong, Q., C. Y. Lee, C. P. Teo, Z. Zheng. 2013. Scheduling arrivals to a stochastic service delivery system using copositive cones. *Oper. Res.* **61**(3): 711–726.

Kong, Q., S. Li, N. Liu, C. P. Teo, Z. Yan. 2020. Appointment scheduling under time-dependent patient no-show behavior. *Management Sci.*

Kouvelis, P., G. Yu. 1997. *Robust Discrete Optimization and Its Applications. Nonconvex Optimization and Its Applications.* Springer, Berlin. US.

Kouvelis, P., A. A. Kurawarwala, G. J. Gutierrez. 1992. Algorithms for robust single and multiple period layout planning for manufacturing systems. *Eur. J. Oper. Res.* **63**(2): 287–303.

Lebedev, V., I. Averbakh. 2006. Complexity of minimizing the total flow time with interval data and minmax regret criterion. *Discrete Appl. Math.* **154**(15): 2167–2177.

Leung, S. C. H., Y. Wu. 2004. A robust optimization model for stochastic aggregate production planning. *Prod. Plan. Control* **15**(5): 502–514.

Levi, R., G. Perakis, J. Uichanco. 2011. Regret Optimization for Stochastic Inventory Models with Spread.

Li, X., K. Natarajan, C. P. Teo, Z. Zheng. 2014. Distributionally robust mixed integer linear programs: Persistency models with applications. *Eur. J. Oper. Res.* **233**(3): 459–473.

Li, B., R. Jiang, J. L. Mathieu. 2019. Ambiguous risk constraints with moment and unimodality information. *Math. Program.* **173**(1–2): 151–192.

Lim, S. 2013. A joint optimal pricing and order quantity model under parameter uncertainty and its practical implementation. *Omega* **41**(6): 998–1007.

Lim, S. 2019. A note on a robust inventory model with stock-dependent demand. *J. Oper. Res. Soc.* **70**(5): 851–866.

Lim, A. E., J. G. Shanthikumar. 2007. Relative entropy, exponential utility, and robust dynamic pricing. *Oper. Res.* **55**(2): 198–214.

Lim, Y. F., C. Wang. 2017. Inventory management based on target-oriented robust optimization. *Management Sci.* **63**(12): 4409–4427.

Lim, A. E., J. G. Shanthikumar, Z. M. Shen. 2006. Model uncertainty, robust optimization, and learning. *Tutorials in Operations Research: Models, Methods, and Applications for Innovative Decision Making* **5**, 66–94.

Liu, S., L. He, Z. J. M. Shen. 2020. Data-driven order assignment for last mile delivery. *Management Sci.*

Liyanage, L. H., J. G. Shanthikumar. 2005. A practical inventory control policy using operational statistics. *Oper. Res. Lett.* **33**(4): 341–348.

Love, D., G. Bayraksan. 2015. Phi-divergence constrained ambiguous stochastic programs for data-driven optimization. Technical report, Department of Integrated Systems Engineering, The Ohio State University, Columbus, Ohio.

Lu, M., L. Ran, Z. J. M. Shen. 2015. Reliable facility location design under uncertain correlated disruptions. *Manuf. Serv. Oper. Manag.* **17**(4): 445–455.

Lu, M., H. Nakao, S. Shen, L. Zhao. 2020. Non-profit resource allocation and service scheduling with cross-subsidization and uncertain resource consumptions. *Omega.* https://doi.org/10.1016/j.omega.2019.102191

Lyu, G., C. P. Teo. (2019) Last mile innovation: The case of the locker alliance network. Available at SSRN 3471166.

Mak, H. Y., Z. J. M. Shen. 2014. Pooling and dependence of demand and yield in multiple-location inventory systems. *Manuf. Serv. Oper. Manag.* **16**(2): 263–269.

Mak, H. Y., Z. J. M. Shen. 2016. Integrated modeling for location analysis. *Foundations and Trends® in Technology, Information and Operations Management* **9**(1–2): 1–152.

Mak, H. Y., Y. Rong, Z. J. M. Shen. 2013. Infrastructure planning for electric vehicles with battery swapping. *Management Sci.* **59**(7): 1557–1575.

Mak, H. Y., Y. Rong, J. Zhang. 2015. Appointment scheduling with limited distributional information. *Management Sci.* **61**(2): 316–334.

Mamani, H., S. Nassiri, M. R. Wagner. 2017. Closed-form solutions for robust inventory management. *Management Sci.* **63**(5): 1625–1643.

Marin, A., J. Salmeron. 2001. A risk function for the stochastic modeling of electric capacity expansion. *Nav. Res. Logist.* **48**(8): 662–683.

Meilijson, I., A. Nádas. 1979. Convex majorization with an application to the length of critical paths. *J. Appl. Probab.* **16**(3): 671–677.

Meng, F., J. Qi, M. Zhang, J. Ang, S. Chu, M. Sim. 2015. A robust optimization model for managing elective admission in a public hospital. *Oper. Res.* **63**(6): 1452–1467.

Mishra, V. K., K. Natarajan, D. Padmanabhan, C. P. Teo, X. Li. 2014. On theoretical and empirical aspects of marginal distribution choice models. *Management Sci.* **60**(6): 1511–1531.

Montemanni, R., J. Barta, M. Mastrolilli, L. M. Gambardella. 2007. The robust traveling salesman problem with interval data. *Transport. Sci.* **41**(3): 366–381.

Moon, I., E. A. Silver. 2000. The multi-item newsvendor problem with a budget constraint and fixed ordering costs. *J. Oper. Res. Soc.* **51**(5): 602–608.

Movahed, K. K., Z. H. Zhang. 2015. Robust design of (s, S) inventory policy parameters in supply chains with demand and lead time uncertainties. *Int. J. Syst. Sci.* **46**(12): 2258–2268.

Mudchanatongsuk, S., F. Ordonez, J. Liu. 2008. Robust solutions for network design under transportation cost and demand uncertainty. *J. Oper. Res. Soc.* **59**(5): 652–662.

Mulvey, J. M., R. J. Vanderbei, S. A. Zenios. 1995. Robust optimization of large-scale systems. *Oper. Res.* **43**(2): 264–281.

Natarajan, K., D. Pachamanova, M. Sim. 2009a. Constructing risk measures from uncertainty sets. *Oper. Res.* **57**(5): 1129–1141.

Natarajan, K., M. Song, C. P. Teo. 2009b. Persistency model and its applications in choice modeling. *Management Sci.* **55**(3): 453–469.

Natarajan, K., C. P. Teo, Z. Zheng. 2011. Mixed 0–1 linear programs under objective uncertainty: A completely positive representation. *Oper. Res.* **59**(3): 713–728.

Natarajan, K., D. Shi, K. C. Toh. 2013. A probabilistic model for minmax regret in combinatorial optimization. *Oper. Res.* **62**(1): 160–181.

Natarajan, K., M. Sim, J. Uichanco. 2017. Asymmetry and ambiguity in newsvendor models. *Management Sci.* **64**(7): 3146–3167.

Ng, T. S., Y. Sun, J. Fowler. 2010. Semiconductor lot allocation using robust optimization. *Eur. J. Oper. Res.* **205**(3): 557–570.

Ng, T. S., J. Fowler, I. Mok. 2012. Robust demand service achievement for the co-production newsvendor. *IIE Trans.* **44**(5): 327–341.

Nie, Y. M., X. Wu, T. Homem-de-Mello. 2012. Optimal path problems with second-order stochastic dominance constraints. *Netw. Spat. Econ* **12**(4): 561–587.

Park, K., K. Lee. 2016. Distribution-robust single-period inventory control problem with multiple unreliable suppliers. *OR Spectrum* **38**(4): 949–966.

Peng, P., L. V. Snyder, A. Lim, Z. Liu. 2011. Reliable logistics networks design with facility disruptions. *Trans Res B: Methodol.* **45**(8): 1190–1211.

Perakis, G., G. Roels. 2008. Regret in the newsvendor model with partial information. *Oper. Res.* **56**(1): 188–203.

Perakis, G., A. Sood. 2006. Competitive multi-period pricing for perishable products: A robust optimization approach. *Math. Program.* **107**(1–2): 295–335.

Pinar, M. C., C. Kizilkale. 2017. Robust screening under ambiguity. *Math. Program.* **163**(1–2): 273–299.

Qi, W., Z. J. M. Shen. 2018. A Smart-City Scope of Operations Management. *Prod. Oper. Manag.* **28**(2): 393–406.

Rahimian, H., S. Mehrotra. 2019. Distributionally robust optimization: A review. arXiv preprint arXiv:1908.05659.

Roy, B. 2010. Robustness in operational research and decision aiding: A multi-faceted issue. *Eur. J. Oper. Res.* **200**(3): 629–638.

Rusmevichientong, P., H. Topaloglu. 2012. Robust assortment optimization in revenue management under the multinomial logit choice model. *Oper. Res.* **60**(4): 865–882.

Scarf, H. E. 1957. A Min-Max Solution of an Inventory Problem. K. J. Arrow, S. Karlin, H. E. Scarf, eds. *Studies in Applied Probability and Management Science*. Stanford University Press, Stanford, CA.

See, C. T., M. Sim. 2010. Robust approximation to multiperiod inventory management. *Oper. Res.* **58**(3): 583–594.

Shaked, M., J. G. Shanthikumar. 1997. Supermodular stochastic orders and positive dependence of random vectors. *J. Multivar. Anal.* **61**(1): 86–101.

Shaked, M., J. G. Shanthikumar. 2007. *Stochastic Orders*, Springer Science & Business Media, Berlin.

Shen, S., M. You, Y. Ma. 2017. Single-commodity stochastic network design under demand and topological uncertainties with insufficient data. *Nav. Res. Logist.* **64**(2): 154–173.

Simchi-Levi, D. 2013. OMforum.OMresearch: From problem-driven to data-driven research. *Manuf. Serv. Oper. Manag.* **16**(1): 2–10.

Snyder, L. V., M. S. Daskin. 2006. Stochastic *p*-robust location problems. *IIE Trans.* **38**(11): 971–985.

Soyster, A. L. 1973. Technical note–convex programming with set-inclusive constraints and applications to inexact linear programming. *Oper. Res.* **21**(5): 1154–1157.

Sözüer S., A. C. Thiele. 2016. The state of robust optimization. *Robustness Analysis in Decision Aiding, Optimization, and Analytics*, International Series in Operations Research and Management Science. Springer, Cham, 89–112.

Sungur, I., F. Ordonez, M. Dessouky. 2008. A robust optimization approach for the capacitated vehicle routing problem with demand uncertainty. *IIE Trans.* **40**(5): 509–523.

Swaminathan, J. M. 2018. Big data analytics for rapid, impactful, sustained, and efficient (RISE) humanitarian operations. *Prod. Oper. Manag.* **27**(9): 1696–1700.

Tang, C. S. 2006. Perspectives in supply chain risk management. *Int. J. Prod. Econ.* **103**(2): 451–488.

Tang, C. S. 2015. The past, present, and future of Manufacturing & Service Operations Management. *Manuf. Serv. Oper. Manag.* **17**(1): 1–3.

Thorsen, A., T. Yao. 2017. Robust inventory control under demand and lead time uncertainty. *Ann. Oper. Res.* **257**(1–2): 207–236.

Vairaktarakis, G. L. 2000a. Robust multi-item newsboy models with a budget constraint. *Int. J. Prod. Econ.* **66**(3): 213–226.

Vairaktarakis, G. L. 2000b. Robust multi-item newsboy models with a budget constraint. *Int. J. Prod. Econ.* **66**(3): 213–226.

Wagner, M. R. 2015. Robust purchasing and information asymmetry in supply chains with a price-only contract. *IIE Trans.* **47**(8): 819–840.

Wagner, M. R. 2018. Robust inventory management: An optimal control approach. *Oper. Res.* **66**(2): 426–447.

Wang, X., J. Zhang. 2015. Process flexibility: A distribution-free bound on the performance of k-chain. *Oper. Res.* **63**(3): 555–571.

Wang, C. X., S. Webster, S. Zhang. 2014. Robust price-setting newsvendor model with interval market size and consumer willingness-to-pay. *Int. J. Prod. Econ.* **154**: 100–112.

Wang, Z., P. W. Glynn, Y. Ye. 2016. Likelihood robust optimization for data-driven problems. *CMS* **13**(2): 241–261.

Wei, C., Y. Li, X. Cai. 2011. Robust optimal policies of production and inventory with uncertain returns and demand. *Int. J. Prod. Econ.* **134**(2): 357—367.

Whitt, W., W. You. 2017. Using robust queueing to expose the impact of dependence in single-server queues. *Oper. Res.* **66** (1): 184–199.

Wiesemann, W., Kuhn D., B. Rustem. 2012a. Multi-resource allocation in stochastic project scheduling. *Ann. Oper. Res.* **193**(1): 193–220.

Wiesemann, W., D. Kuhn, B. Rustem. 2012b. Robust resource allocations in temporal networks. *Math. Program.* **135**(1–2): 437–471.

Wiesemann, W., D. Kuhn, M. Sim. 2014. Distributionally robust convex optimization. *Oper. Res.* **62**(6): 1358–1376.

Xie, W., S. Ahmed. 2018. Distributionally robust chance constrained optimal power flow with renewables: A conic reformulation. *IEEE Trans. Power Syst.* **33**(2): 1860–1867.

Xu, H., C. Caramanis, S. Mannor. 2009. Robust regression and lasso. D. Koller, D. Schuurmans, Y. Bengio, L. Bottou, eds., *Advances in Neural Information Processing Systems 21*, Curran Associates, Inc., Red Hook, NY, 1801–1808.

Yan, Z., C. Cheng, K. Natarajan, C. P. Teo. 2016. Marginal Estimation Price Optimization': Data Driven Multi-Product Pricing Problem. SSRN Scholarly Paper ID 2832385, Social Science Research Network, Rochester, NY.

Yan, Z., S. Y. Gao, C. P. Teo. 2017. On the design of sparse but efficient structures in operations. *Management Sci.* **64**(7): 3421–3445.

Yanıkoğlu I., D. den Hertog. 2013. Safe approximations of ambiguous chance constraints using historical data. *INFORMS J. Comput.* **25**(4): 666–681.

Yanıkoğlu I., B. Gorissen, D. den Hertog, 2017. Adjustable Robust Optimization–A Survey and Tutorial. Available online at ResearchGate.

Yu, G. 1997. Robust economic order quantity models. *Eur. J. Oper. Res.* **100**(3): 482–493.

Yu, G., J. Yang. 1998. On the robust shortest path problem. *Comput. Oper. Res.* **25**(6): 457–468.

Yue, J., B. Chen, M. C. Wang. 2006. Expected value of distribution information for the newsvendor problem. *Oper. Res.* **54**(6): 1128–1136.

Zanjani, M. K., D. Ait-Kadi, M. Nourelfath. 2010. Robust production planning in a manufacturing environment with random yield: A case in sawmill production planning. *Eur. J. Oper. Res.* **201**(3): 882–891.

Zeng, B., L. Zhao. 2013. Solving two-stage robust optimization problems using a column-and-constraint generation method. *Oper. Res. Lett.* **41**(5): 457–461.

Zhang, M. 2011. Two-stage minimax regret robust uncapacitated lot-sizing problems with demand uncertainty. *Oper. Res. Lett.* **39**(5): 342–345.

Zhang, Y., Z. J. M. Shen, S. Song. 2016. Distributionally robust optimization of two-stage lot-sizing problems. *Prod. Oper. Manag.* **25**(12): 2116–2131.

Zhang, Y., S. Shen, S. Erdogan. 2017a. Solving 0–1 Semidefinite Programs for Distributionally Robust Allocation of Surgery Blocks. SSRN Scholarly Paper ID 2997524, Social Science Research Network, Rochester, NY.

Zhang, Y., S. Shen, S. A. Erdogan. 2017b. Distributionally robust appointment scheduling with moment-based ambiguity set. *Oper. Res. Lett.* **45**(2): 139–144.

Zhang, Y., S. Shen, J. L. Mathieu. 2017c. Distributionally robust chance-constrained optimal power flow with uncertain renewables and uncertain reserves provided by loads. *IEEE Trans. Power Syst.* **32**(2): 1378–1388.

Zhang, Y., Z. J. M. Shen, S. Song. 2018a. Exact algorithms for distributionally β-robust machine scheduling with uncertain processing times. *INFORMS J. Comput.* **30**(4): 662–676.

Zhang, Y., S. Song, Z. M. Shen, C. Wu. 2018b. Robust shortest path problem with distributional uncertainty. *IEEE Trans. Intell. Transp. Syst.* **19**(4): 1080–1090.

Zhao, M., N. Freeman. 2016. Optimal Sourcing with Ambiguously Correlated Suppliers. SSRN Scholarly Paper ID 2851228, Social Science Research Network, Rochester, NY.

Zhao, M., N. K. Freeman. 2018. Robust sourcing from suppliers under ambiguously correlated major disruption risks. *Prod. Oper. Manag.* **28**(2): 441–456.

Zhu, Z., J. Zhang, Y. Ye. 2013. Newsvendor optimization with limited distribution information. *Optim. Methods Softw.* **28**(3): 640–667.